

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Introduction to Differential Calculus

Week 5 Day 2 AM



COURSE SURVEY

The SSILP Mathematics Survey is now open until 10 Aug. 2026 (11.59pm), you should have received an email on this. Please follow the link provided in the email to complete the survey. Your feedback is very important for us to improve the course.



OUTLINE

① Tangent Lines and Derivatives

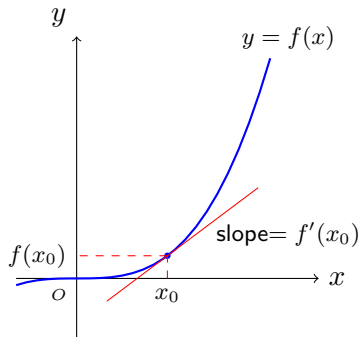
② Rules of Differentiation

③ Instantaneous Rates of Change

④ Summary

GEOMETRICAL MEANING OF DERIVATIVES

Geometrically, the derivative $f'(x_0)$ of a function f at the point x_0 is the slope/gradient of the tangent line to the graph of $f(x)$ at $x = x_0$.



How can we formally define this? We use the slope of secant lines.
Try this [Geogebra applet](#)!

DEFINITION OF DERIVATIVES

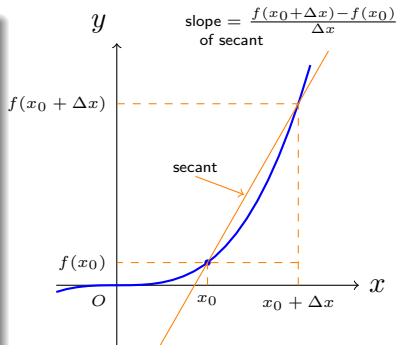
DEFINITION:

The **derivative of f at x_0** is

$$\frac{df}{dx}(x_0) := \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$$

or (writing $x = x_0 + \Delta x$)

$$\frac{df}{dx}(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0}$$



whenever the **limit exists**, in which case we say that f is **differentiable** at x_0 . If the limit doesn't exist, then f is **not differentiable** at x_0 . We also denote the derivative as $f'(x_0)$ or $\left. \frac{df}{dx} \right|_{x=x_0}$ or $\left. \frac{dy}{dx} \right|_{x=x_0}$ if $y = f(x)$. Note that Δx can be **positive** or **negative**!

DERIVATIVE OF $f(x) = x$

Remark: Quite often, if the point x_0 is arbitrary, we'll drop the subscript and just use the point x .

E.g.: We can compute the derivative of $f(x) = x$ by definition as follows:

$$\begin{aligned}f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{x + \Delta x - x}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} 1 = 1.\end{aligned}$$

So

$$f'(x) = 1.$$

DERIVATIVE OF $f(x) = x^2$

E.g.: We can compute the derivative of $f(x) = x^2$ by definition as follows. It is also common to use h instead of Δx to lighten the notation.

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\&= \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \\&= \lim_{h \rightarrow 0} \frac{2xh + h^2}{h} \\&= \lim_{h \rightarrow 0} (2x + h) = 2x.\end{aligned}$$

So

$$f'(x) = 2x.$$

DERIVATIVE OF $f(x) = x^3$

E.g.: We can compute the derivative of $f(x) = x^3$ by definition as follows.

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) \\ &= 3x^2. \end{aligned}$$

So

$$f'(x) = 3x^2.$$

DIFFERENTIATION AND TANGENT LINE

Remark: Since at those x such that f is differentiable, $f'(x)$ gives the (unique) value of the derivative of f at x , so $f'(x)$ is a function of x (possibly with a smaller domain than f). We'll call this function f' the **derivative of f** and denote it as f' or $f'(x)$ or $\frac{df}{dx}$ or $\frac{d}{dx}f(x)$.

DEFINITION:

The process of computing $f'(x)$ from $f(x)$ is called **differentiation**. The verb is **to differentiate**.

E.g.: We know from the previous slide that, if we differentiate x^3 , we'll get $3x^2$, i.e. $3x^2$ is the derivative of x^3 .

DEFINITION:

If $f(x)$ is differentiable at x_0 , then the **tangent line** (to $y = f(x)$) at x_0 is the straight line with slope $f'(x_0)$ that passes through the point $(x_0, f(x_0))$. Thus, the equation of the tangent line is

$$y = f'(x_0)(x - x_0) + f(x_0).$$

ACTIVITY 1

By using the limit definition of the derivative, find the derivatives of the following functions.

(a) $f(x) = x^n$ where n is a positive integer.

Hint: Binomial theorem.

(b) $f(x) = \frac{1}{x}$ for $x \neq 0$.

(c) $f(x) = \sqrt{x}$ for $x > 0$.

ACTIVITY 1: SOLUTION

(a)

$$\begin{aligned}f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^n - x^n}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{x^n + nx^{n-1}\Delta x + \binom{n}{2}x^{n-2}(\Delta x)^2 + \binom{n}{3}x^{n-3}(\Delta x)^3 + \cdots + (\Delta x)^n - x^n}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} (nx^{n-1} + \binom{n}{2}x^{n-2}\Delta x + \binom{n}{3}x^{n-3}(\Delta x)^2 + \cdots + (\Delta x)^{n-1}) \\&= nx^{n-1}.\end{aligned}$$

This derivative formula in fact holds for **any** $n \in \mathbb{R}$: integer, positive, negative, rational or irrational number.

ACTIVITY 1: SOLUTION

(b)

$$\begin{aligned}f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \left(\frac{\frac{1}{x + \Delta x} - \frac{1}{x}}{\Delta x} \right) \\&= \lim_{\Delta x \rightarrow 0} \frac{-\Delta x}{(x + \Delta x)x\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{-1}{(x + \Delta x)x} \\&= -\frac{1}{x^2}.\end{aligned}$$

Notice that this result coincides with the case of $n = -1$ in the previous formula $\frac{d}{dx}x^n = nx^{n-1}$ by writing $\frac{1}{x} = x^{-1}$.

ACTIVITY 1: SOLUTION

(c)

$$\begin{aligned}f'(x) &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x} \\&= \lim_{\Delta x \rightarrow 0} \left(\frac{\sqrt{x + \Delta x} - \sqrt{x}}{\Delta x} \times \frac{\sqrt{x + \Delta x} + \sqrt{x}}{\sqrt{x + \Delta x} + \sqrt{x}} \right) \\&= \lim_{\Delta x \rightarrow 0} \frac{x + \Delta x - x}{\Delta x(\sqrt{x + \Delta x} + \sqrt{x})} \\&= \lim_{\Delta x \rightarrow 0} \frac{1}{\sqrt{x + \Delta x} + \sqrt{x}} = \frac{1}{2\sqrt{x}}.\end{aligned}$$

Notice that this result coincides with the case of $n = \frac{1}{2}$ in the previous formula $\frac{d}{dx}x^n = nx^{n-1}$ by writing $\sqrt{x} = x^{\frac{1}{2}}$.

OUTLINE

① Tangent Lines and Derivatives

② Rules of Differentiation

③ Instantaneous Rates of Change

④ Summary

RULES OF DIFFERENTIATION

It is cumbersome and tedious to use the limit definition of the derivative every time we compute the derivative of a function.

You will learn some rules of computing derivatives in this section. They are more efficient and faster in computing derivatives than the limit definition.

However, these rules are limited to certain cases and you should go back to the definition when the rules are not applicable.

All the rules are essentially proven by using the limit definition of the derivative.

RULES OF DIFFERENTIATION

SUM RULE

$$\frac{d}{dx}(f(x) + g(x)) = f'(x) + g'(x).$$

E.g.:

$$\frac{d}{dx}(x + x^2 + x^3) = 1 + 2x + 3x^2.$$

CONSTANT MULTIPLE RULE

$$\frac{d}{dx}(c \cdot f(x)) = c \left(\frac{d}{dx} f(x) \right) = c \cdot f'(x).$$

E.g.:

$$\frac{d}{dx}(7x^2) = 7(2x) = 14x.$$

POWER RULE

POWER RULE

$$\frac{d}{dx}x^n = nx^{n-1} \quad \text{for all constant exponents } n \in \mathbb{R}.$$

Remark: Note that the power rule is true for all exponent $n \in \mathbb{R}$, which is more general than what we have proven in activity 1. We'll skip the proof for now.

$$\text{E.g.: } \frac{d}{dx}\sqrt[3]{x} = \frac{d}{dx}x^{\frac{1}{3}} = \frac{1}{3}x^{\frac{1}{3}-1} = \frac{1}{3}x^{-\frac{2}{3}} = \frac{1}{3x^{\frac{2}{3}}}.$$

$$\text{E.g.: } \frac{d}{dx}\frac{1}{\sqrt{x}} = \frac{d}{dx}x^{-\frac{1}{2}} = -\frac{1}{2}x^{-\frac{1}{2}-1} = -\frac{1}{2}x^{-\frac{3}{2}} = -\frac{1}{2x^{\frac{3}{2}}}.$$

$$\text{E.g.: } \frac{d}{dx}x^\pi = \pi x^{\pi-1}.$$

E.g.: $\frac{d}{dx}x^x \neq x \cdot x^{x-1} = x^x$, because the exponent x is not a constant.

PRODUCT RULE

PRODUCT RULE

$$\frac{d}{dx}(f(x) \cdot g(x)) = f'(x) \cdot g(x) + f(x) \cdot g'(x).$$

E.g.: Given that $\frac{d}{dx} \tan x = \sec^2 x$. If $f(x) = x \cdot \tan x$, then

$$f'(x) = 1 \cdot \tan x + x \cdot \frac{d}{dx} \tan x = \tan x + x \sec^2 x.$$

E.g.: Given that $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$. If $f(x) = \sin(x) \cos(x)$, then

$$\begin{aligned} f'(x) &= \cos(x) \cos(x) + \sin(x)(-\sin(x)) \\ &= \cos^2(x) - \sin^2(x) \\ &= \cos(2x). \end{aligned}$$

QUOTIENT RULE

QUOTIENT RULE

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g(x)^2}.$$

E.g.: Given that $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$. If $f(x) = \tan(x) = \frac{\sin(x)}{\cos(x)}$, then

$$f'(x) = \frac{\cos(x) \cos(x) - \sin(x)(-\sin(x))}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x).$$

E.g.: If $f(x) = \frac{x^2+1}{x^3+1}$, then

$$\begin{aligned} f'(x) &= \frac{(2x)(x^3+1) - (x^2+1)(3x^2)}{(x^3+1)^2} \\ &= \frac{2x^4 + 2x - (3x^4 + 3x^2)}{(x^3+1)^2} \\ &= \frac{-x^4 - 3x^2 + 2x}{(x^3+1)^2}. \end{aligned}$$

CHAIN RULE

CHAIN RULE

$$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x).$$

Let $u = g(x)$ and $y = f(g(x)) = f(u)$, then

CHAIN RULE (ALTERNATIVE NOTATION)

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$

E.g.: To compute $\frac{d}{dx}(3x+2)^2$, let $u = 3x+2$ and $y = (3x+2)^2 = u^2$. By the chain rule,

$$\begin{aligned}\frac{d}{dx}(3x+2)^2 &= \frac{dy}{du} \times \frac{du}{dx} \\ &= \frac{d}{du}(u^2) \times \frac{d}{dx}(3x+2) \\ &= 2u \times 3 = 2(3x+2) \times 3 = 6(3x+2).\end{aligned}$$

CHAIN RULE - EXAMPLE

E.g.: Given that $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$, then to compute $\frac{d}{dx} \sin(\cos(x))$, let $u = \cos x$ and $y = \sin(\cos(x)) = \sin u$. Then the chain rule gives

$$\begin{aligned}\frac{d}{dx} \sin(\cos(x)) &= \frac{dy}{du} \times \frac{du}{dx} \\ &= \frac{d}{du} (\sin u) \times \frac{d}{dx} (\cos x) \\ &= \cos u \times (-\sin x) \\ &= \cos(\cos(x)) \times (-\sin x) = -\sin(x) \cos(\cos(x)).\end{aligned}$$

If you are comfortable to apply the chain rule directly, then you can write

$$\frac{d}{dx} \sin(\cos(x)) = \cos(\cos(x)) \cdot (-\sin(x)) = -\sin(x) \cos(\cos(x)).$$

ACTIVITY 2

Differentiate the following functions with respect to x .

(a) $x^{\frac{4}{3}} + \frac{1}{x^5}$

(b) $\frac{1}{\sqrt[4]{3x}}$

(c) $(x^3 + 1) \sin(x^2)$, given that $\frac{d}{dx} \sin x = \cos x$.

(d) $\frac{1}{\sqrt{x^2 + 4x}}$

(e) $\sqrt{\frac{2-x}{3x+4}}$

ACTIVITY 2: SOLUTION

(a)

$$\frac{d}{dx} \left(x^{\frac{4}{3}} + \frac{1}{x^5} \right) = \frac{4}{3} x^{\frac{1}{3}} - \frac{5}{x^6}.$$

(b)

$$\begin{aligned} \frac{d}{dx} \left(\frac{1}{\sqrt[4]{3x}} \right) &= \frac{d}{dx} (3x)^{-\frac{1}{4}} \\ &= -\frac{1}{4} (3x)^{-\frac{1}{4}-1} \cdot 3 \\ &= -\frac{3}{4} \frac{1}{\sqrt[4]{(3x)^5}}. \end{aligned}$$

(c)

$$\begin{aligned} \frac{d}{dx} \left((x^3 + 1) \sin(x^2) \right) &= 3x^2 \sin(x^2) + (x^3 + 1) \cos(x^2)(2x) \\ &= 3x^2 \sin(x^2) + 2x(x^3 + 1) \cos(x^2). \end{aligned}$$

ACTIVITY 2: SOLUTION

(d)

$$\begin{aligned}\left(\frac{1}{\sqrt{x^2 + 4x}}\right)' &= \left((x^2 + 4x)^{-\frac{1}{2}}\right)' = -\frac{1}{2}(x^2 + 4x)^{-\frac{3}{2}}(2x + 4) \\ &= -(x + 2)(x^2 + 4x)^{-\frac{3}{2}}.\end{aligned}$$

(e)

$$\begin{aligned}\left(\sqrt{\frac{2-x}{3x+4}}\right)' &= \left(\left(\frac{2-x}{3x+4}\right)^{\frac{1}{2}}\right)' = \frac{1}{2}\left(\frac{2-x}{3x+4}\right)^{-\frac{1}{2}}\left(\frac{2-x}{3x+4}\right)' \\ &= \frac{1}{2}\left(\frac{3x+4}{2-x}\right)^{\frac{1}{2}} \frac{-1 \cdot (3x+4) - 3(2-x)}{(3x+4)^2} \\ &= \frac{1}{2}\left(\frac{3x+4}{2-x}\right)^{\frac{1}{2}} \frac{-10}{(3x+4)^2} = \frac{-5}{\sqrt{(2-x)(3x+4)^3}}.\end{aligned}$$

OUTLINE

① Tangent Lines and Derivatives

② Rules of Differentiation

③ Instantaneous Rates of Change

④ Summary

PHYSICAL MEANING OF DERIVATIVES

Physically,

$$f'(x_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\text{change in } f(x) \text{ at } x = x_0}{\text{change in } x \text{ at } x = x_0}$$

is the **instantaneous rate of change of $f(x)$ w.r.t. x at x_0** .

If $f'(x_0) > 0$, then $f(x)$ will increase when x increases from $x = x_0$.

If $f'(x_0) < 0$, then $f(x)$ will decrease when x increases from $x = x_0$.

In practice, we have the approximate estimation (the smaller the change in x , the better the estimation):

$$f'(x_0) \approx \frac{\text{change in } f(x) \text{ at } x = x_0}{\text{change in } x \text{ at } x = x_0}$$

$$\Rightarrow (\text{change in } f(x) \text{ at } x = x_0) \approx f'(x_0) \cdot (\text{change in } x \text{ at } x = x_0).$$

PHYSICAL MEANING OF DERIVATIVES - EXAMPLE

E.g.: The cost C (in dollars) of building a house A square feet in area is given by $C = f(A)$. Then knowing

$$f'(1000) = 2000 \text{ dollars/sq ft}$$

tells us that when $A = 1000$ sq ft, the house requires a cost of approximately 2000 dollars for every square foot to be built. We get an approximation because the cost rate of 2000 dollars/sq ft is instantaneous, i.e. the increment from $A = 1000$ sq ft should be infinitesimally small for the rate 2000 dollars/sq ft, an increment of one square foot is large. If the increment from $A = 1000$ sq ft is 0.05 sq ft, then the approximate change in C is more accurate, which is

$$f'(1000) \cdot (0.05) = 2000 \times 0.05 = 100 \text{ dollars,}$$

i.e. the cost of building a house of 1000.05 sq ft is roughly $f(1000) + 100$ dollars.

PHYSICAL MEANING OF DERIVATIVES - EXAMPLE

E.g.: The cost of extracting T tons of ore from a copper mine is $C = f(T)$ dollars. If

$$f'(2000) = 100 \text{ dollars/tons,}$$

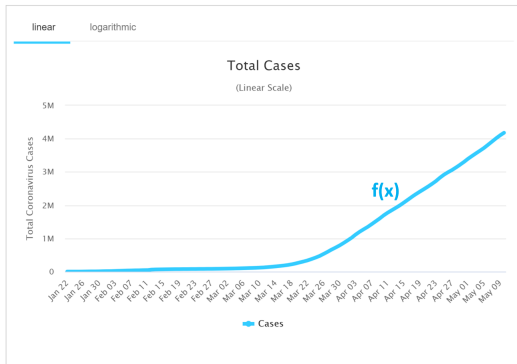
it means that when 2000 tons of ore have already been extracted from the mine, the cost of extracting the next ton is approximately \$ 100.

E.g.: If $q = f(p)$ gives the number of pounds of sugar produced when the price per pound is p dollars, then

$$f'(3) = 50 \text{ lb/\$}$$

means that the instantaneous change of q with respect to p is 50 when $p = 3$, i.e. when the price is \$3 per pound, the quantity produced is increasing at 50 lb/\$. Thus, if the price increased by a dollar, the quantity produced would increase by approximately 50lb.

ACTIVITY 3

Source: Worldometer - www.worldometers.info 

Total covid-19 cases during 22 Jan 2021 to 9 May 2021.

Determine the units and interpret the following statements:

(a) $f(\text{Apr } 17) = 2,249,004$.

(b) $f'(\text{Apr } 17) = 76,929$.

ACTIVITY 3: SOLUTION

- (a) $f(\text{Apr } 17) = 2,249,004$ cases. On 17 April 2021, the total number of covid-19 cases worldwide was 2,249,004.
- (b) $f'(\text{Apr } 17) = 76,929$ cases/day. The number of total cases on 17 April 2021 was increasing at a rate of 76,929 cases/day. So we can expect the number of new cases on 18 April 2021 to be approximately $2,249,004 + 76,929 = 2,325,933$.

ACTIVITY 4

The time for a chemical reaction, T (in minutes), is a function of the amount of catalyst present, a (in milliliters), so $T = f(a)$.

- (a) If $f(5) = 18$, what are the units of 5? What are the units of 18? What does this statement tell us about the reaction?
- (b) If $f'(5) = -3$, what are the units of 5? What are the units of -3 ? What does this statement tell us?

ACTIVITY 4: SOLUTION

- (a) The statement $f(5) = 18$ means that when 5 milliliters of catalyst are present, the reaction will take 18 minutes. Thus, the units for 5 are ml while the units for 18 are minutes.
- (b) As in part (a), 5 is measured in ml. Since f' tells how fast T changes per unit a , we have f' measured in minutes/ml. If the amount of catalyst increases by 1 ml from 5 ml, the reaction time decreases approximately by about 3 minutes.

RELATED RATES OF CHANGE

The 2nd notation of the chain rule $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$ is often used to solve related rates of change problems.

E.g.: A spherical balloon is inflating at a rate of $2 \text{ cm}^3/\text{s}$. What is the time rate of change of its radius r , when $r = 10 \text{ cm}$?

If we denote the time at t (in s) and let its volume be V (in cm^3), then it's given that $\frac{dV}{dt} = 2 \text{ cm}^3/\text{s}$ and we are asked to find $\frac{dr}{dt}$.

We use the chain rule to relate these two quantities:

$$\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}.$$

We can find $\frac{dV}{dr} = 4\pi r^2$ from the relationship $V = \frac{4}{3}\pi r^3$. With this, the chain rule becomes

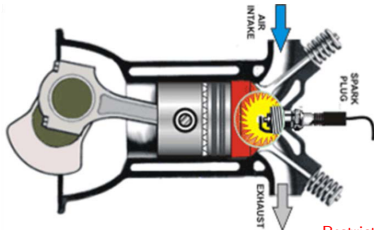
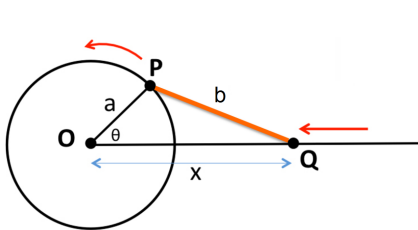
$$2 = \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt} \quad \Rightarrow \quad \frac{dr}{dt} = \frac{1}{2\pi r^2}.$$

At $r = 10 \text{ cm}$, we have $\frac{dr}{dt} = \frac{1}{2\pi \times 10^2} = \frac{1}{200\pi} \text{ cm/s}$.

ACTIVITY 5

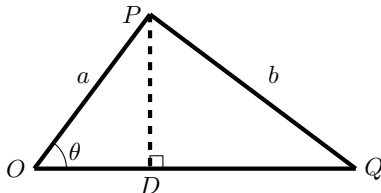
Let's model the movement of the piston of a 4-stroke engine. The metal bar of length b has one end attached at the point P to a circle of radius a . Point Q at the other end can slide back and forth along the x -axis.

- (a) Find x as a function of θ .
- (b) Assume the lengths are in cm, and the angular speed $\frac{d\theta}{dt}$ is 2 radians per second counterclockwise. Find the velocity in terms of a at which the point Q is moving when $\theta = \frac{\pi}{2}$. Use the fact that $\frac{d}{dx} \sin x = \cos x$ and $\frac{d}{dx} \cos x = -\sin x$.



ACTIVITY 5: SOLUTION

(a)



From the triangle OPQ :

$$\frac{OD}{a} = \cos \theta \Rightarrow OD = a \cos \theta,$$
$$\frac{PD}{a} = \sin \theta \Rightarrow PD = a \sin \theta.$$

From the triangle PQD , we get: $PD^2 + DQ^2 = b^2$.

So, $a^2 \sin^2 \theta + DQ^2 = b^2 \Rightarrow DQ = \sqrt{b^2 - a^2 \sin^2 \theta}$. Thus,

$$x = OD + DQ = a \cos \theta + \sqrt{b^2 - a^2 \sin^2 \theta}.$$

ACTIVITY 5: SOLUTION

- (b) We are given $\frac{d\theta}{dt} = 2$ and asked to find $\frac{dx}{dt}$. They are related by the chain rule:

$$\frac{dx}{dt} = \frac{dx}{d\theta} \cdot \frac{d\theta}{dt} = 2 \cdot \frac{dx}{d\theta}.$$

We use the result $x = a \cos \theta + \sqrt{b^2 - a^2 \sin^2 \theta}$ from part (a) to find $\frac{dx}{d\theta}$ by differentiating x with respect to θ (a and b are constant):

$$\begin{aligned} \frac{dx}{d\theta} &= -a \sin \theta + \frac{1}{2} \frac{(-2a^2 \sin \theta \cos \theta)}{\sqrt{b^2 - a^2 \sin^2 \theta}} \\ &= -a \sin \theta - \frac{a^2 \sin \theta \cos \theta}{\sqrt{b^2 - a^2 \sin^2 \theta}}. \end{aligned}$$

Substituting for $\theta = \frac{\pi}{2}$,

$$\left. \frac{dx}{dt} \right|_{\theta=\frac{\pi}{2}} = 2 \left. \frac{dx}{d\theta} \right|_{\theta=\frac{\pi}{2}} = -2a \sin \frac{\pi}{2} - 2 \frac{a^2 \sin \frac{\pi}{2} \cos \frac{\pi}{2}}{\sqrt{b^2 - a^2 \sin^2 \frac{\pi}{2}}} = -2a \text{ cm/s}.$$

OUTLINE

① Tangent Lines and Derivatives

② Rules of Differentiation

③ Instantaneous Rates of Change

④ Summary

SUMMARY

- Tangent Lines and Derivatives
 - geometrical meaning of derivatives.
 - definition of derivatives and tangent line
- Rules of Differentiation
 - sum rule, constant multiple rule, power rule, product rule, quotient rule, chain rule.
 - the chain rule and related rates of change.
- Instantaneous Rates of Change
 - physical meaning of derivatives as instantaneous rates of change

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) section 13.3.